Bearings

Exam code: 0607

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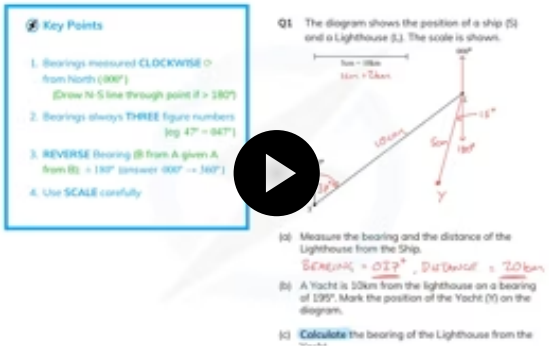

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Bearings

Help me understand

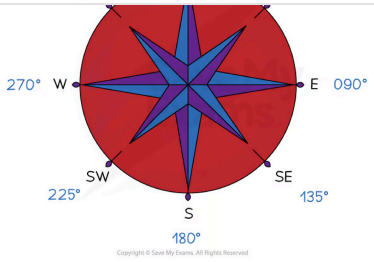
What are bearings?

- **Bearings** are a way of describing an **angle**
  - They are commonly used in **navigation**
- There are **three rules** which must be followed when using a bearing:
  - They are measured **from North**
    - North is usually straight up on a scale drawing or map, and should be labelled on the diagram
  - They are measured **clockwise**
  - The angle should always be written with **3 digits**
    - 059° instead of just 59°
- Knowing the **compass directions** and their respective bearings can be helpful

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



Bearings of compass points

How do I find a bearing between two points?

- Identify where you need to **start**
  - "The bearing **of A from B**" means **start at B** and find the bearing to A
  - "The bearing **of B from A**" means **start at A** and find the bearing to B
- Draw a **North line** at the **starting point**
- Draw a **line between the two points**
- **Measure** the angle between the **North line** and the **line joining the points**
  - Measure **clockwise** from North
- Write the angle using **3 figures**

How do I draw a point on a bearing?

- You might be asked to plot a point that is a **given distance** from another point and on a **given bearing**
- **STEP 1**
  Draw a **North line** at the point you wish to measure the bearing **from**
  - If you are given the bearing **from A to B** draw the North line at **A**
- **STEP 2**
  Measure the **angle** of the bearing given **from the North line** in the **clockwise direction**
- **STEP 3**
  Draw a line and add the **point B** at the **given distance**

How do I find the bearing of B from A if I know the bearing of A from B?

- If the **bearing of A from B** is **less than 180°**
  - **Add 180°** to it to find the **bearing of B from A**
- If the **bearing of A from B** is **more than 180°**
  - **Subtract 180°** from it to find the **bearing of B from A**

How do I answer trickier questions involving bearings?

- Bearings questions may involve the use of **Pythagoras** or **trigonometry** to find missing distances (lengths) and directions (angles)

Examiner Tips and Tricks

Make sure you have all the **equipment** you need for your maths exams.

A rubber and pencil sharpener can be essential as these questions are all about accuracy. Make sure you can see and read the markings on your ruler and protractor.

Examiner Tips and Tricks

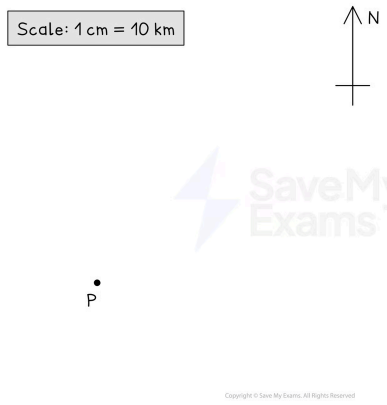
Always draw a big, clear diagram and annotate it, be especially careful to label the angles in the correct places!

Worked Example

A ship sets sail from the point P, as shown on the map below.

It sails on a bearing of 105° until it reaches the point Q, 70 km away. The ship then changes path and sails on a bearing of 065° for a further 35 km, where its journey finishes.

Show on the map below the point Q and the final position of the ship.



Answer:

Draw in a north line at the point P

Measure an angle of 105° clockwise from the north line

Making sure you are accurate, carefully make a small but visible mark on the map

Draw a line from P through the mark you have made. Make this line long so that you can easily measure along it accurately

Use the scale given on the map (1 cm = 10 km) to work out the number of cm that would represent 70 km

70 km = 70 ÷ 10 = 7 cm

Accurately measure 7 cm from the point P along the line and make a clear mark on the line
Label this point Q

A bearing of 065 means 65° clockwise from the North

First, draw a North line at the point Q, then carefully measure an angle of 65° clockwise from this line. Make a mark and then draw a line from Q through this mark

35 km = 35 ÷ 10 = 3.5 cm

Accurately measure 3.5 cm from the point Q along this new line and make a clear mark on the line
This is the final position of the ship.

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